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Subject Name : Machine Learning

Unit No: 4

Unit Name : Support Vector Machine

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Unit No.: 4.1

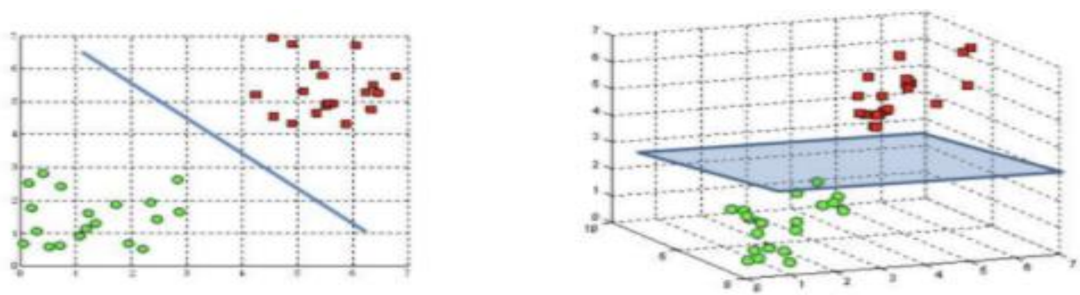
Faculty Name :

Unit No:1

Lecture No: 1

Hyperplane

- Hyperplanes are decision boundaries that help classify the data points.
- Data points falling on either side of the hyperplane can be attributed to different classes.
- Also, the dimension of the hyperplane depends upon the number of features.
- If the number of input features is 2, then the hyperplane is just a line.
- If the number of input features is 3, then the hyperplane becomes a two-dimensional plane.



Hyperplanes in 2D and 3D feature space

Linear Separability

- **Linear separability in machine learning refers to the ability to classify data points into two distinct classes using a single straight line (in 2D), plane (in 3D), or hyperplane (in n-dimensions).**
- If a straight boundary can perfectly separate two classes, the data is linearly separable; if not, it is nonlinearly separable.
- Two sets of data points are linearly separable if they can be separated by a linear decision surface (hyperplane), meaning all points of one class lie on one side of the surface, and all points of the other class lie on the other side.
- **Linear Models:** Algorithms such as Linear Regression, Logistic Regression, Naive Bayes, and Perceptrons, SVM with Linear Kernel work well on linearly separable data.

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Non-linearly separable data

- Non-linearly separable data is described as data that cannot be separated using a simple straight line (requires a complex classifier).
- **Non-Linear Models**: When data is not linearly separable, following models are needed to create complex, non-linear boundaries.
  - Neural Networks with hidden layers,
  - Support Vector Machines (SVM) with kernels,
  - decision trees,
  - Random Forest, and
  - K-Nearest Neighbours (KNN)
- Non-linear classification refers to categorizing those instances that are not linearly separable.

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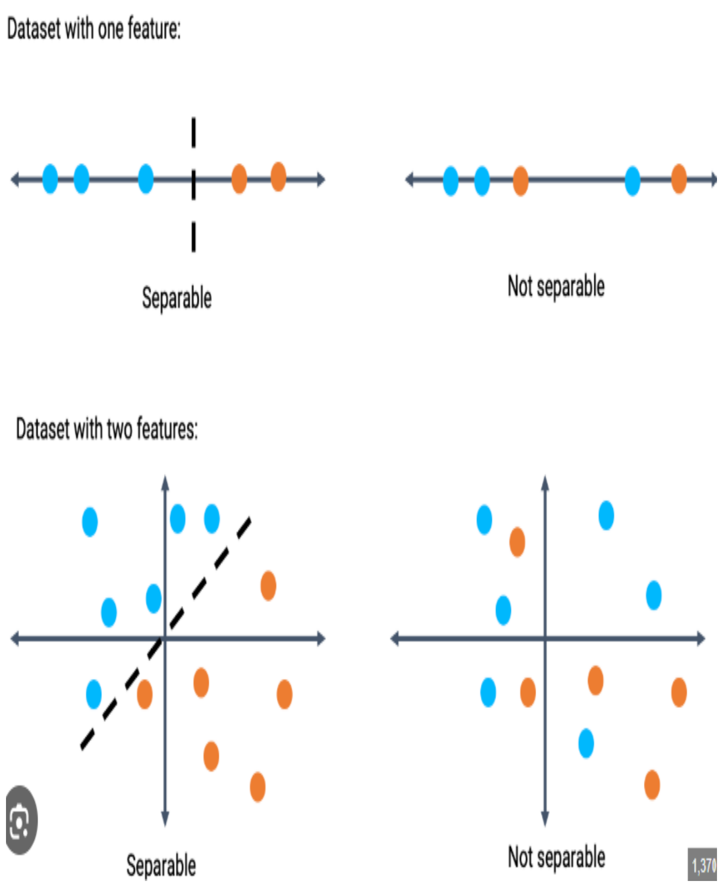
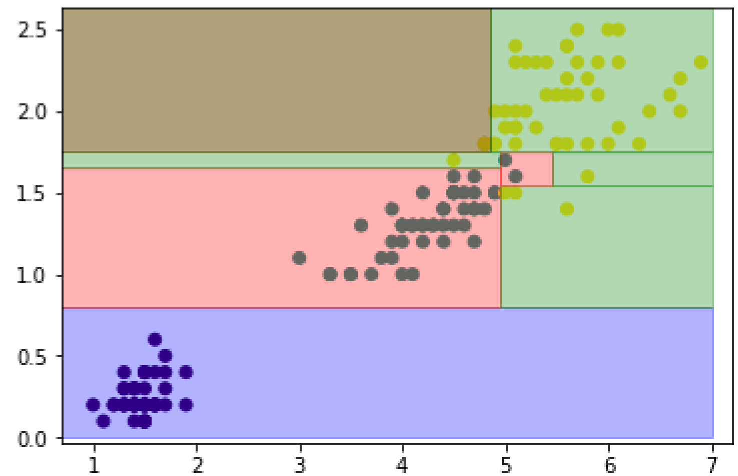
Examples of Linearly and Non Linearly Separable Data Set

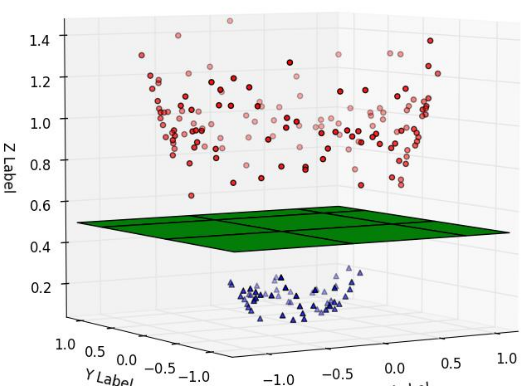


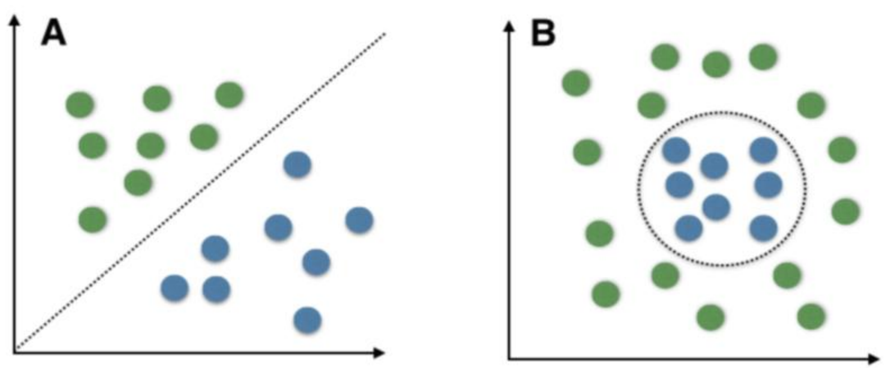


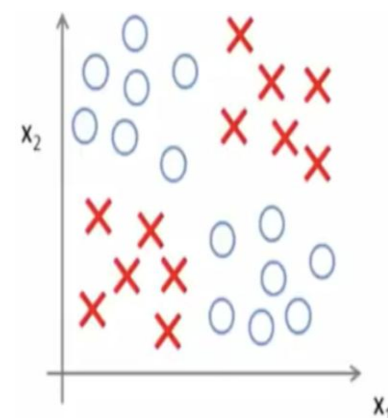
Non-Linearly Separable Data Set

How to Identify Linear Separability

Visual Inspection: Plot the data in 2D or 3D to see if a straight line/plane can divide the groups.







A: Linearly Separable Data B: Non-Linearly Separable Data

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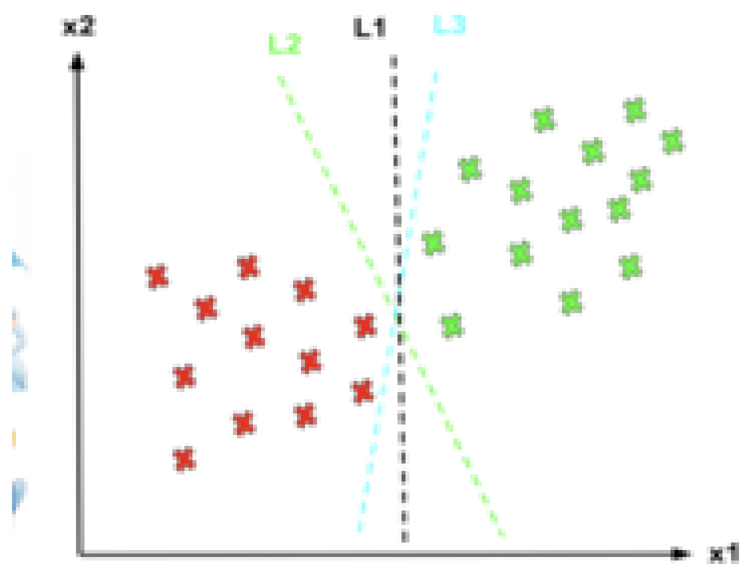
Introduction to Support Vector Machine (SVM)

- "Support Vector Machine" (SVM) is a supervised machine learning algorithm which can be used for both **classification or regression challenges.**
- However, it is **mostly used in classification problems.**
- In the SVM algorithm, we plot each data item as a point in n-dimensional space (where n is number of features you have) with the value of each feature being the value of a particular coordinate.
- Then, we perform classification by finding the hyper-plane that differentiates the two classes very well.

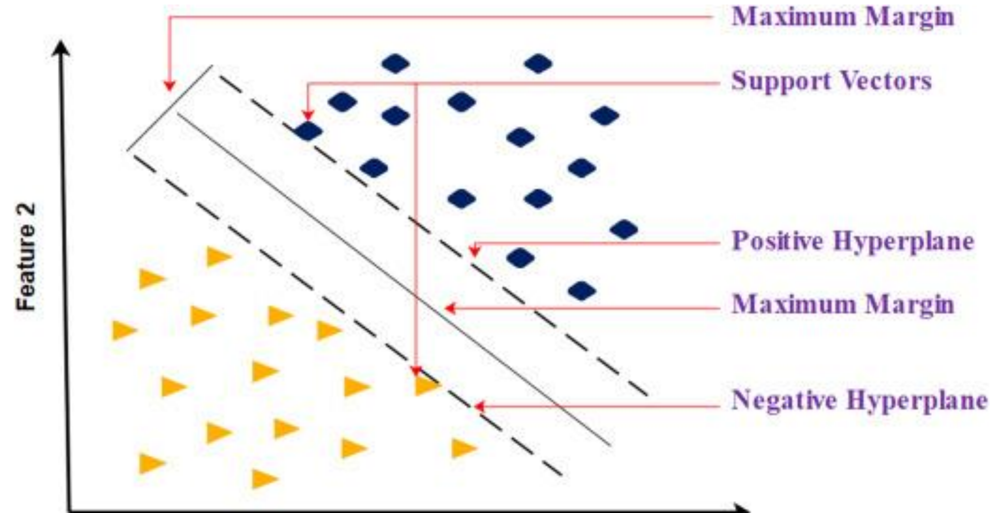
SVM

- Imagine the labelled training set are two classes of data points.
- To separate the two classes, there are so many possible options of hyperplanes that separate correctly. As shown in the graph below, we can achieve exactly the same result using different hyperplanes (L1, L2, L3).
- However, if we add new data points, the consequence of using various hyperplanes will be very different in terms of classifying new data point into the right group of class.



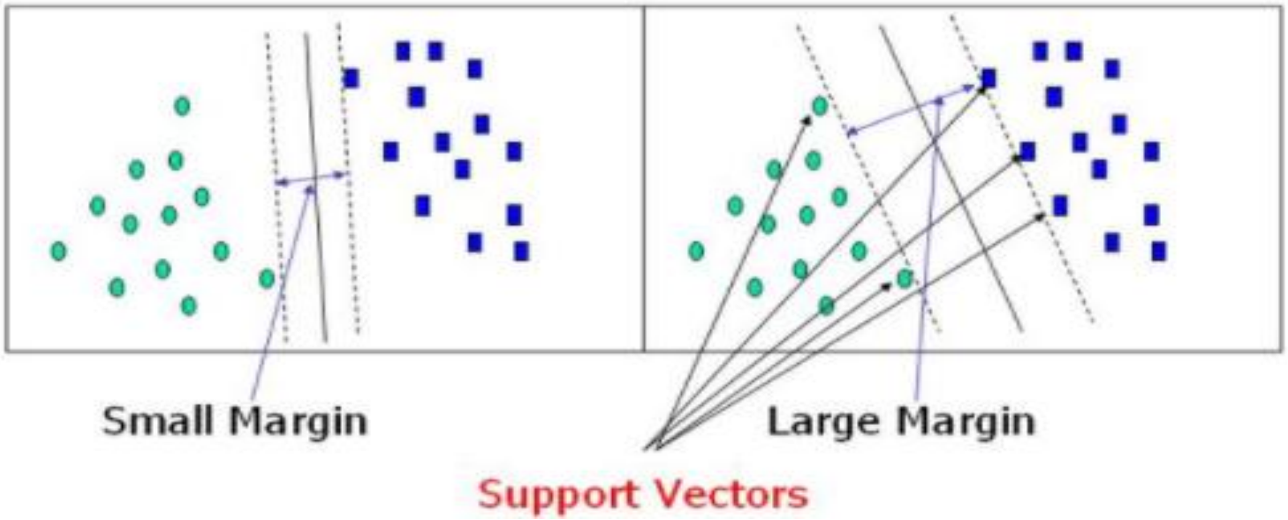
How can we decide a separating line for the classes? Which hyperplane shall we use?

- To separate the two classes of data points, there are many possible hyperplanes that could be chosen.
- **Hyperplane**: A decision boundary separating different classes in feature space and is represented by the equation $wx + b = 0$ in linear classification.
- **Margin** is the distance between the decision boundary (hyperplane) and the closest data points from each class, known as **support vectors.**
- **Optimal Hyperplane:** Our objective is to find a plane that has the maximum margin, i.e the maximum distance between data points of both classes.



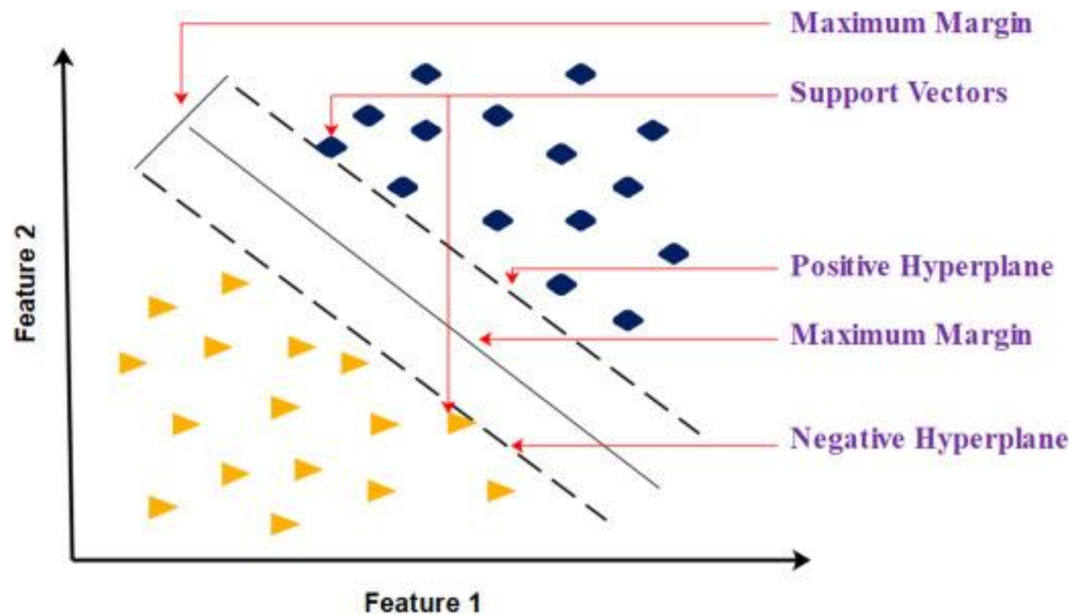
Main Goal of SVM

- The main goal of SVM is to maximize the margin between the two classes. The larger the margin the better the model performs on new and unseen data. Thus, it reduces the generalisation error
- Maximizing the margin distance provides some reinforcement so that future data points can be classified with more confidence.
- The decision boundaries with smaller margins usually lead to overfitting.



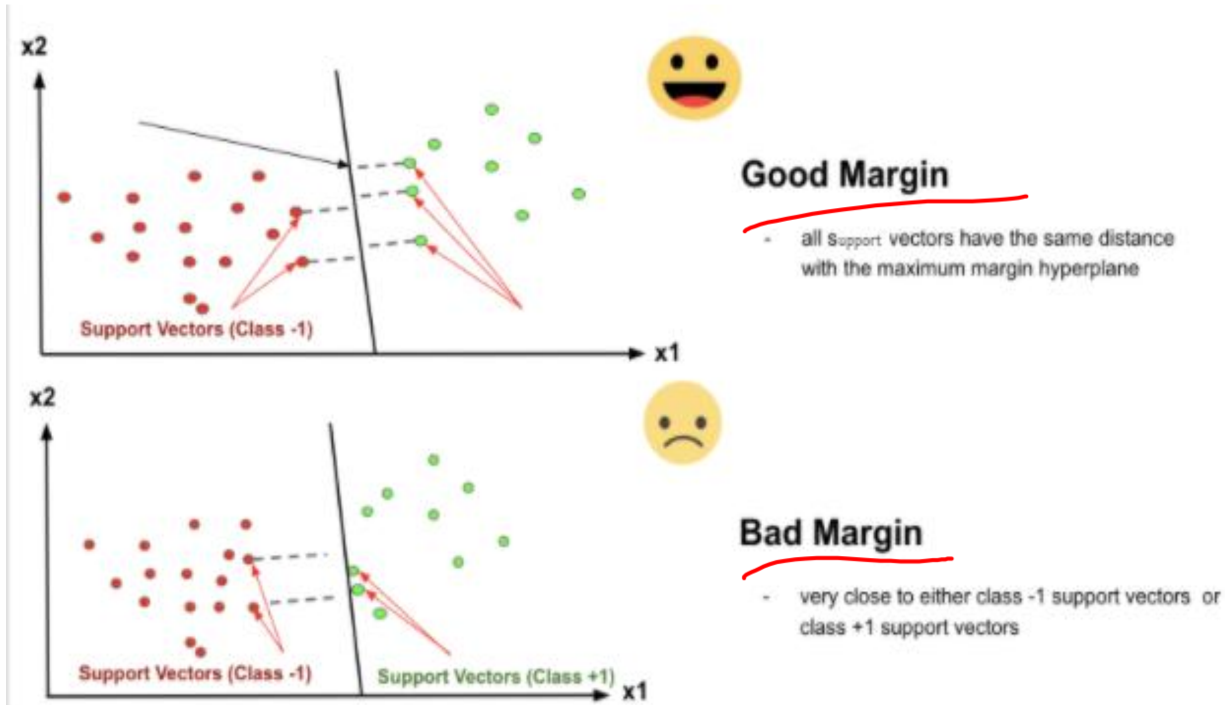
Support Vector

- **Support vectors** are data points that are closest to the hyperplane and influence the position and orientation of the hyperplane.
- Using these support vectors, we maximize the margin of the classifier.
- Deleting the support vectors will change the position of the hyperplane. These are the points that help us build our SVM.
- In other words, only support vectors points contributes to the result of the algorithm, and other points are not.
- If a data point is not a support vector, removing it has no effect on the model. On the other hand, deleting the support vectors will then change the position of the hyperplane.



Margin

- The distance of the support vectors from the hyperplane is called the ***margin,*** which is a separation of a line to the closest class points.
- We would like to choose a hyperplane that maximises the margin between classes. The graph below shows what good margin and bad margin are.



Hard Margin and Soft Margin

- **Hard Margin:** If the training data is perfectly linearly separable, we can select two parallel hyperplanes that separate the two classes of data, so that the distance between them is as large as possible. So ,**Hard Margin** is a maximum-margin hyperplane that perfectly separates the data without misclassifications.
- **Soft Margin:** As most of the real-world data are not fully linearly separable, we will allow some margin violation to occur, which is called soft margin classification. It is better to have a large margin, even though some constraints are violated. Margin violation means choosing a hyperplane, which can allow some data points to stay in either the incorrect side of the hyperplane and between the margin and the correct side of the hyperplane.
- Soft Margin allows some misclassifications by introducing slack variables, balancing margin maximization and misclassification penalties when data is not perfectly separable.